Lecture 14: Bonding in organic molecules-1

Jin Xie (谢琎), SPST, ShanghaiTech

Email: xiejin@shanghaitech.edu.cn

Classroom: Teaching Center 401

Wednesday/Friday: 10:15-11:00; 11:10-11:55

2025/11/28

Major points of last lecture

- **Quantum mechanics and molecular structure**

    12. Summary and Comparison of the LCAO and Valence Bond Methods
    13. Predicting Molecular Structures and Shapes
    14. Using the LCAO and Valence Bond Methods Together
    15. Summary and Comparison of the LCAO and Valence Bond Methods

Summary of hybridization results

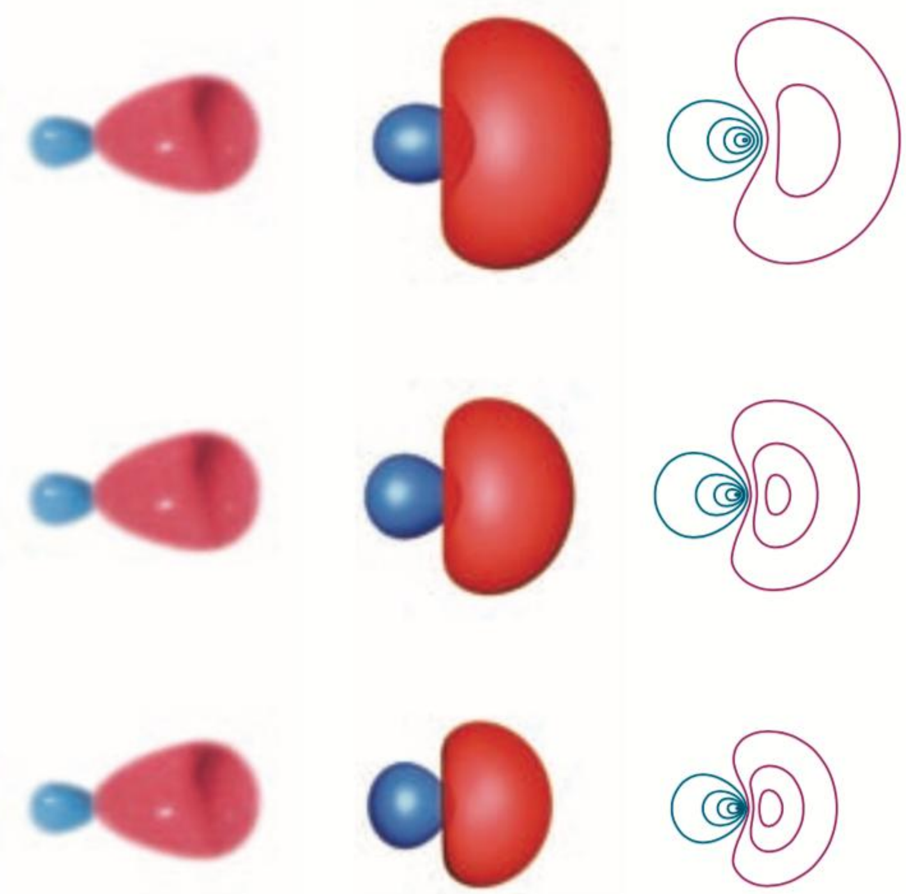
A close relationship between the VSEPR theory and the hybrid orbital approach:

- **steric numbers of 2, 3, and 4 corresponding to sp, sp^2, and sp^3 hybridization**

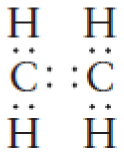
TABLE 6.4

Orbital Hybridization and Molecular Geometry

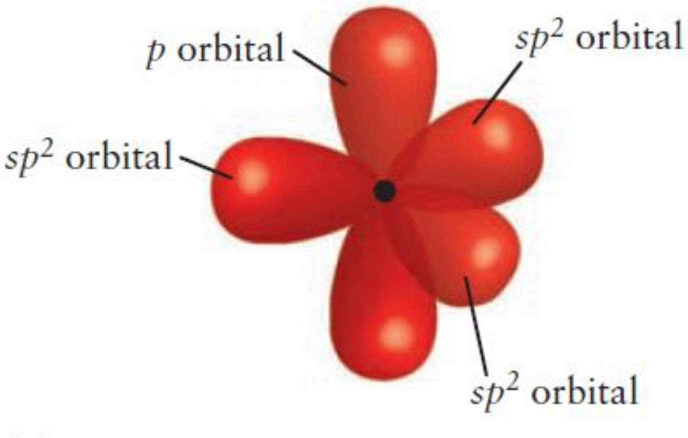
Molecule	Hybrid orbitals on central atom	Molecular geometry	Example
AX_2	*sp*	Linear	BeH_2
AX_3	sp^2	Trigonal planar	BH_3
AX_4	sp^3	Tetrahedral	CH_4



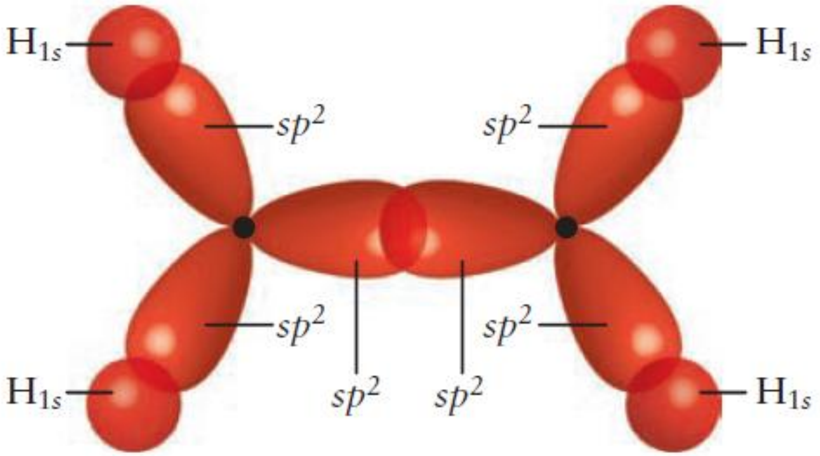
Hybridization and multiple bonds in organic carbon compounds



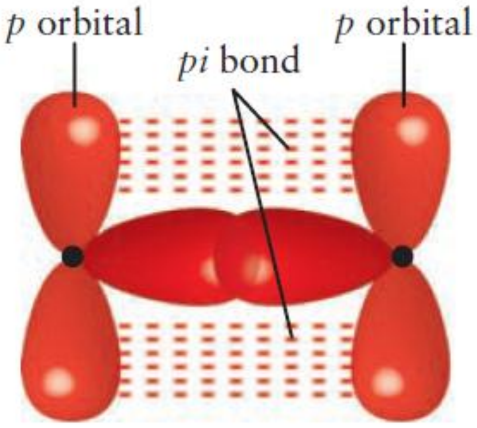
Lewis diagram for C_2H_4.



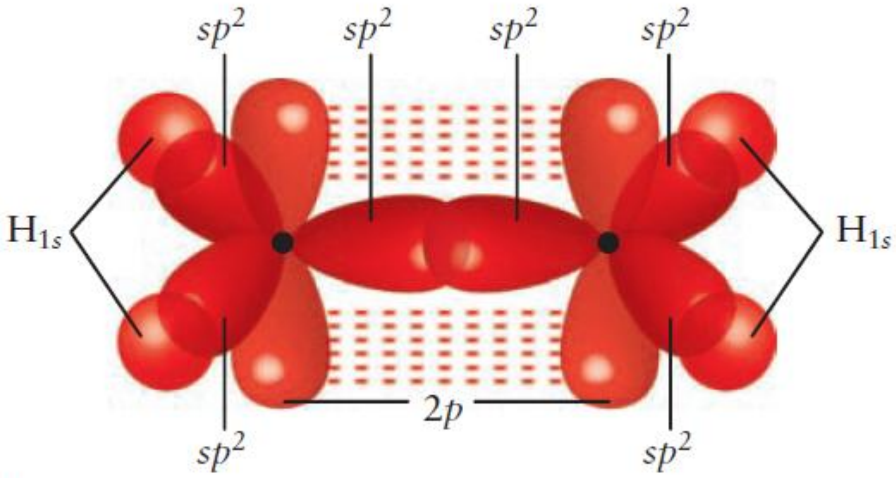
(a)



(b)



(a)

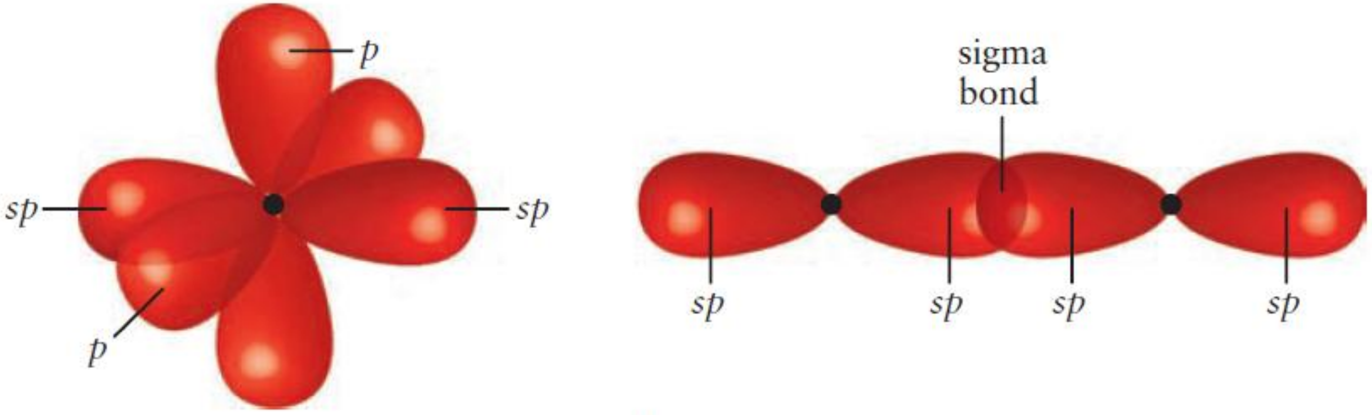


(b)

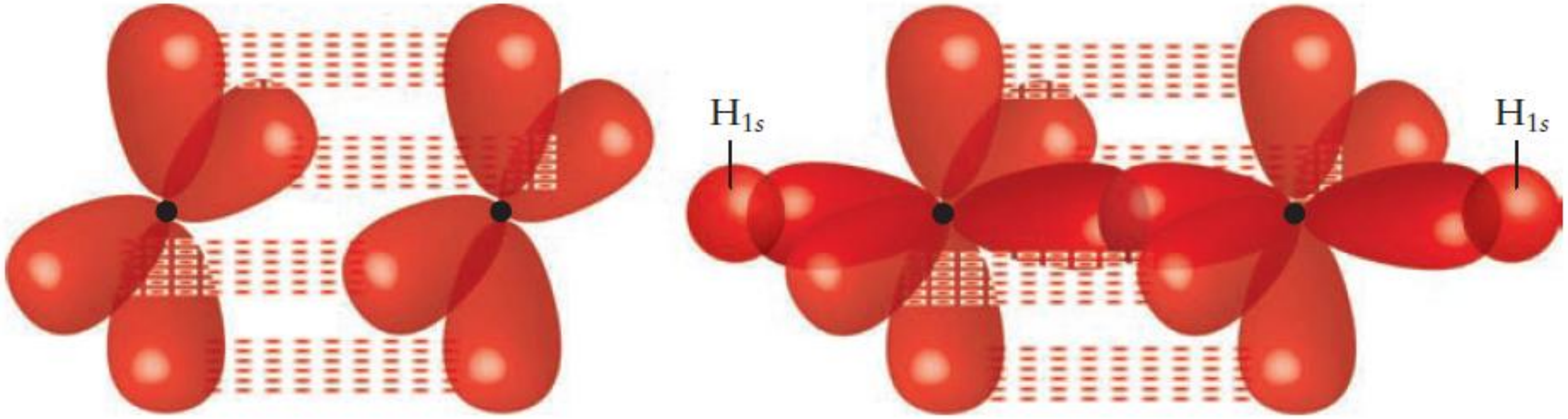
Hybridization and multiple bonds in organic carbon compounds

H:C: : :C:H

Lewis diagram for C_2H_2.



(a) (b)

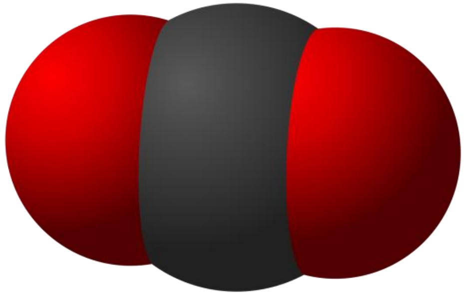


(a) (b)

Predicting Molecular Structures and Shapes

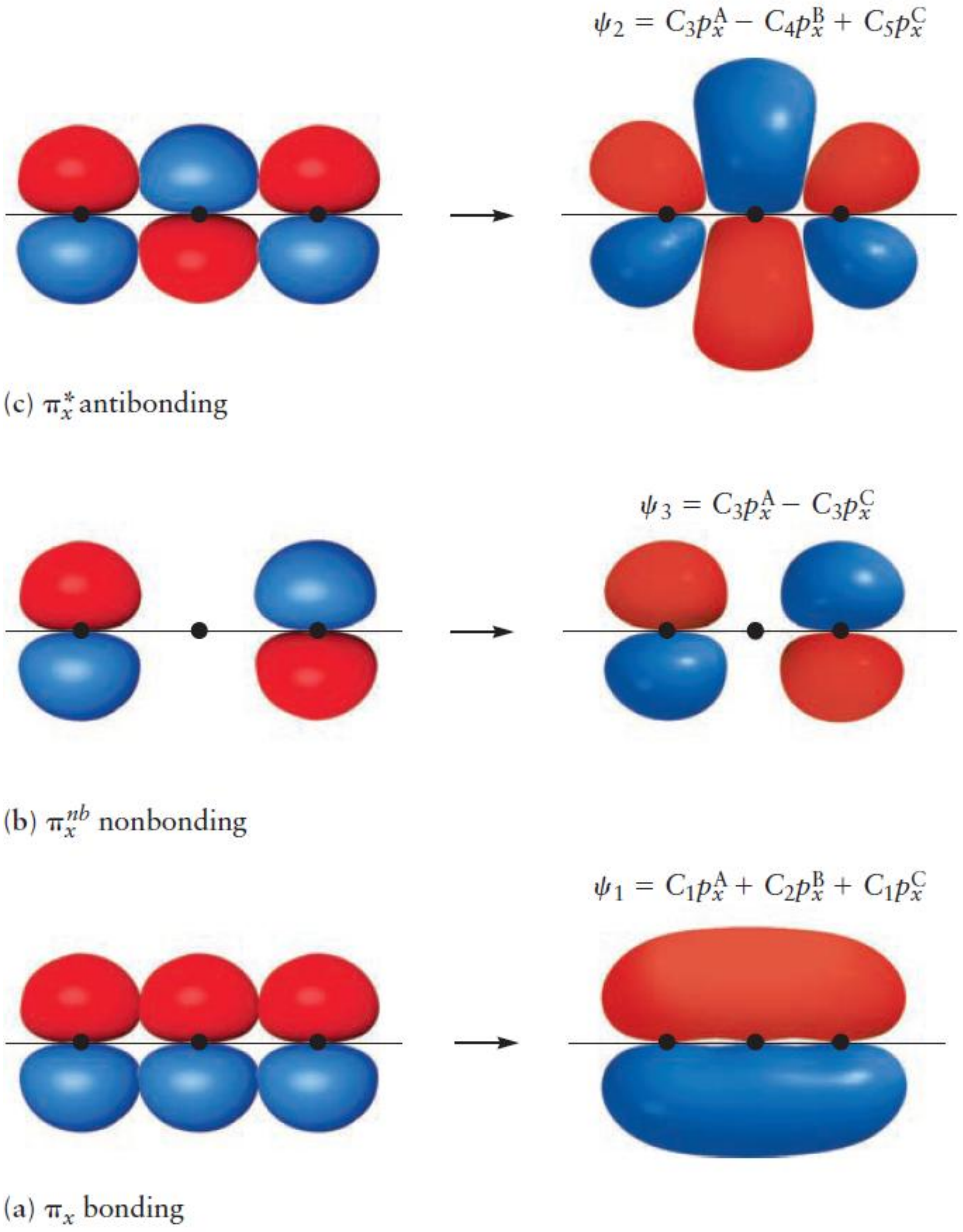
- Determine the empirical formula: an elemental analysis based on combustion.
- Determine the molecular formula: its behavior as a gas or by mass spectrometry.
- Determine the structural formula: write a Lewis diagram based on the molecular formula.
- Determine the molecular shape experimentally, e.g., by X-ray scattering, or predict the shape by using VSEPR theory.
- Identify the hybridization scheme that best explains the shape predicted by VSEPR.

Linear Triatomic Nonhydrides

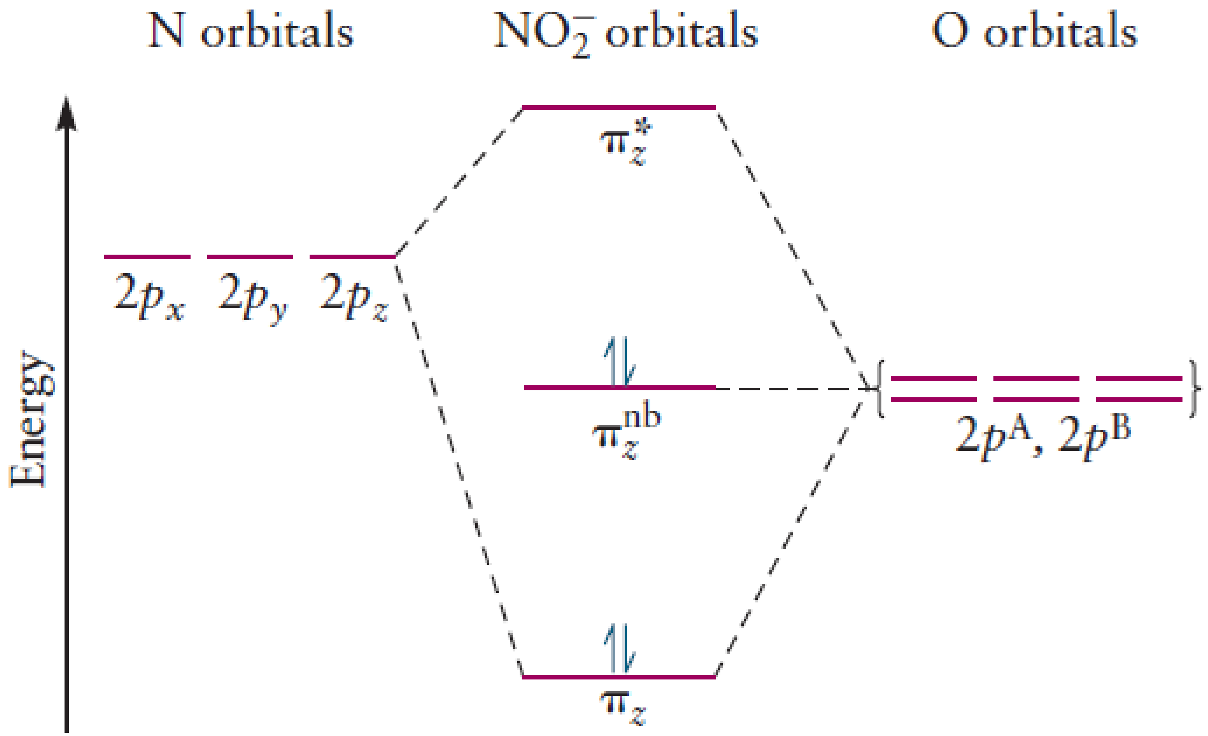


$$O\overset{\pi}{\underset{\sigma}{=}}C\overset{\pi}{\underset{\sigma}{=}}O$$

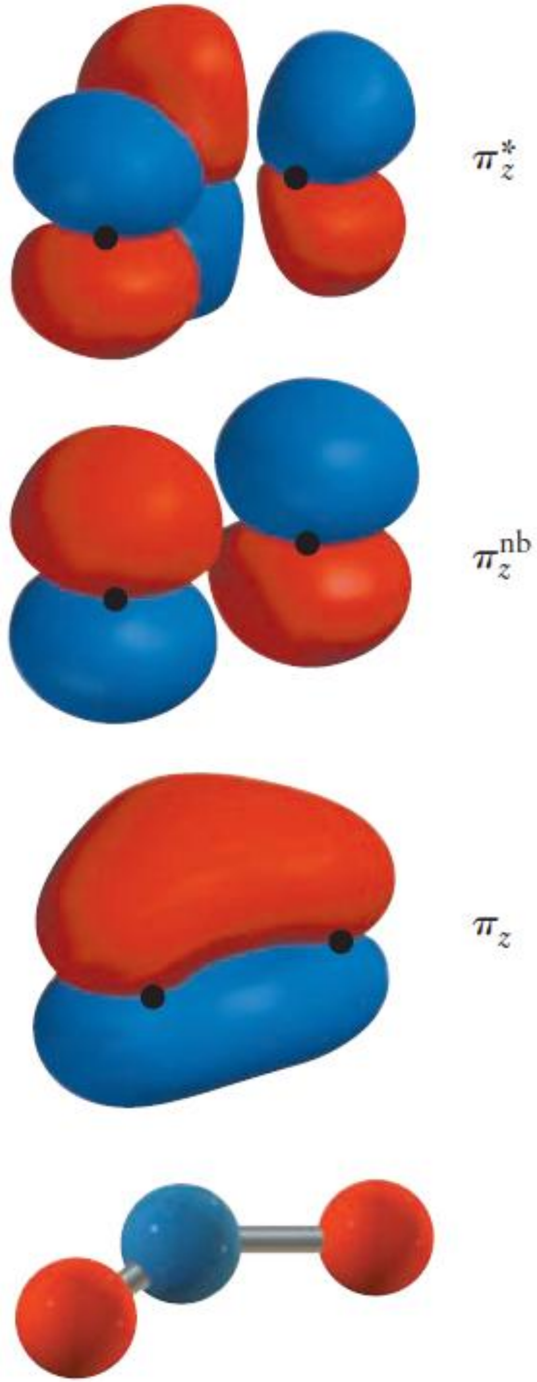
$$E_n = \frac{n^2h^2}{8mL^2} \qquad n = 1, 2, 3, \ldots$$



Nonlinear Triatomic Molecules



(b)



Summary and Comparison of the LCAO and Valence Bond Methods

LCAO: constructs MOs that are delocalized over the entire molecule (linear combinations of the AOs)

H_2

$$\sigma_{g1s} = C_g(R_{AB})\left[\varphi_{1s}^{A} + \varphi_{1s}^{B}\right] \qquad \sigma_{g1s} = \left[1s^{A} + 1s^{B}\right]$$

$$\psi_{MO}^{el} = \sigma_{g1s}(1)\sigma_{g1s}(2) = [1s^{A}(1) + 1s^{B}(1)][1s^{A}(2) + 1s^{B}(2)]$$

$$\psi_{MO}^{el} = [1s^{A}(1)1s^{B}(2) + 1s^{A}(2)1s^{B}(1)] + [1s^{A}(1)1s^{A}(2) + 1s^{B}(1)1s^{B}(2)]$$

VB: a quantum mechanical description of the localized chemical bond between two atoms

H_2

$$\psi_{VB}^{el}(r_{1A}, r_{2B}) = c_1\varphi_A(r_{1A})\varphi_B(r_{2B}) + c_2\varphi_A(r_{2A})\varphi_B(r_{1B})$$

$$\psi_{VB}^{el} = 1s^{A}(1)1s^{B}(2) + 1s^{A}(2)1s^{B}(1)$$

It assumes a good approximation to the molecular electronic wave function for H_2 is the product of an H(1s) orbital centered on atom A, occupied by electron 1, and another H(1s) orbital centered on atom B, occupied by electron 2.

Summary and Comparison of the LCAO and Valence Bond Methods

LCAO (H_2)

$$\psi^{el}_{MO} = [1s^A(1)1s^B(2) + 1s^A(2)1s^B(1)] + [1s^A(1)1s^A(2) + 1s^B(1)1s^B(2)]$$

VB (H_2)

$$\psi^{el}_{VB} = 1s^A(1)1s^B(2) + 1s^A(2)1s^B(1)$$

- **The 1st term in ψ^{el}_{MO} is identical to ψ^{el}_{VB}.**
- **The 2nd term may be labeled ψ_{ionic} because it is a mixture of the ionic states $H_A^-H_B^+$ and $H_A^+H_B^-$**

- **Both methods can be refined to provide more accurate wave functions for molecules and solids.**
- **Both methods have been improved significantly.**

one approach for improving ψ_{VB}

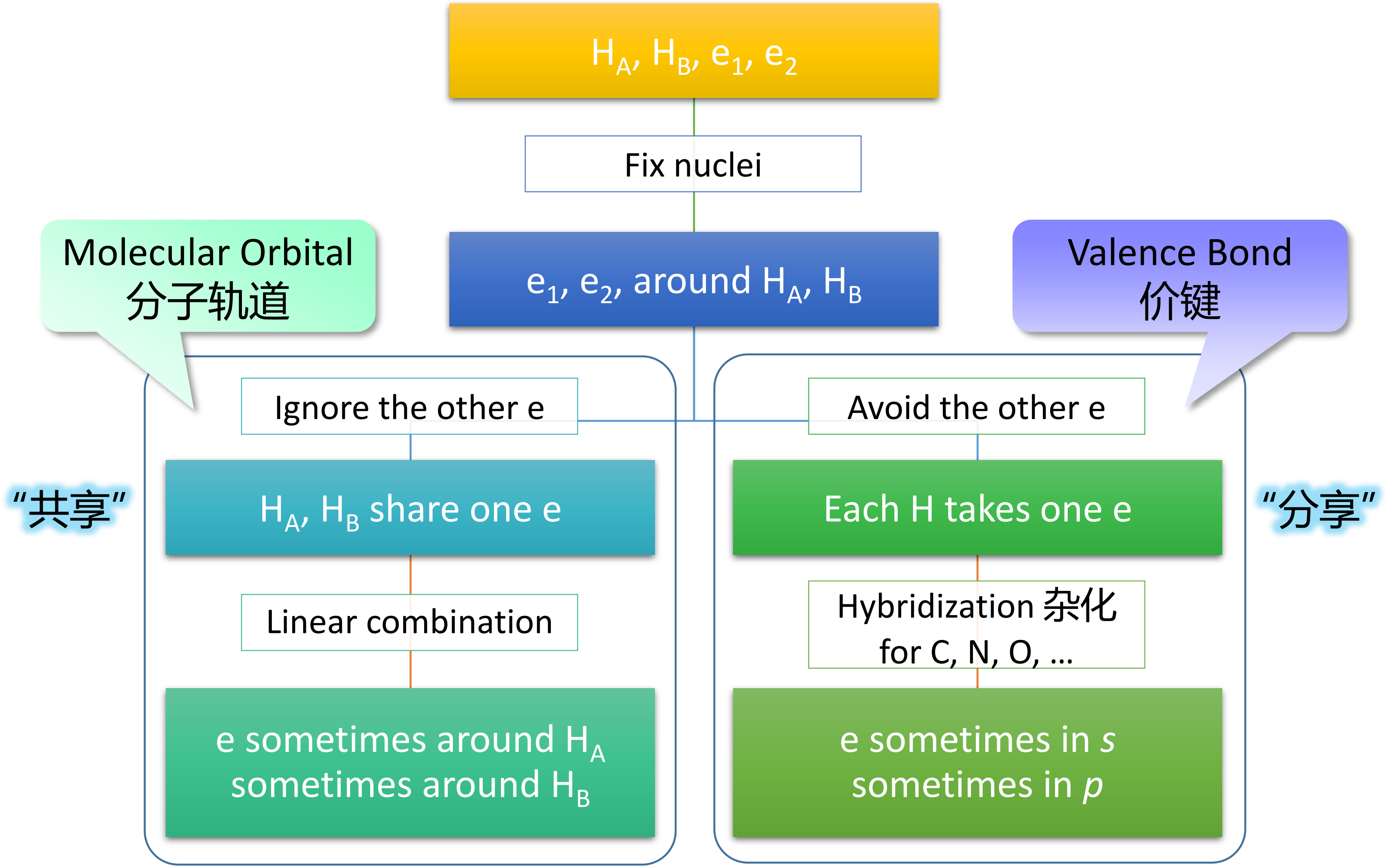
$$\psi_{improved} = \psi_{VB} + \lambda\psi_{ionic}$$

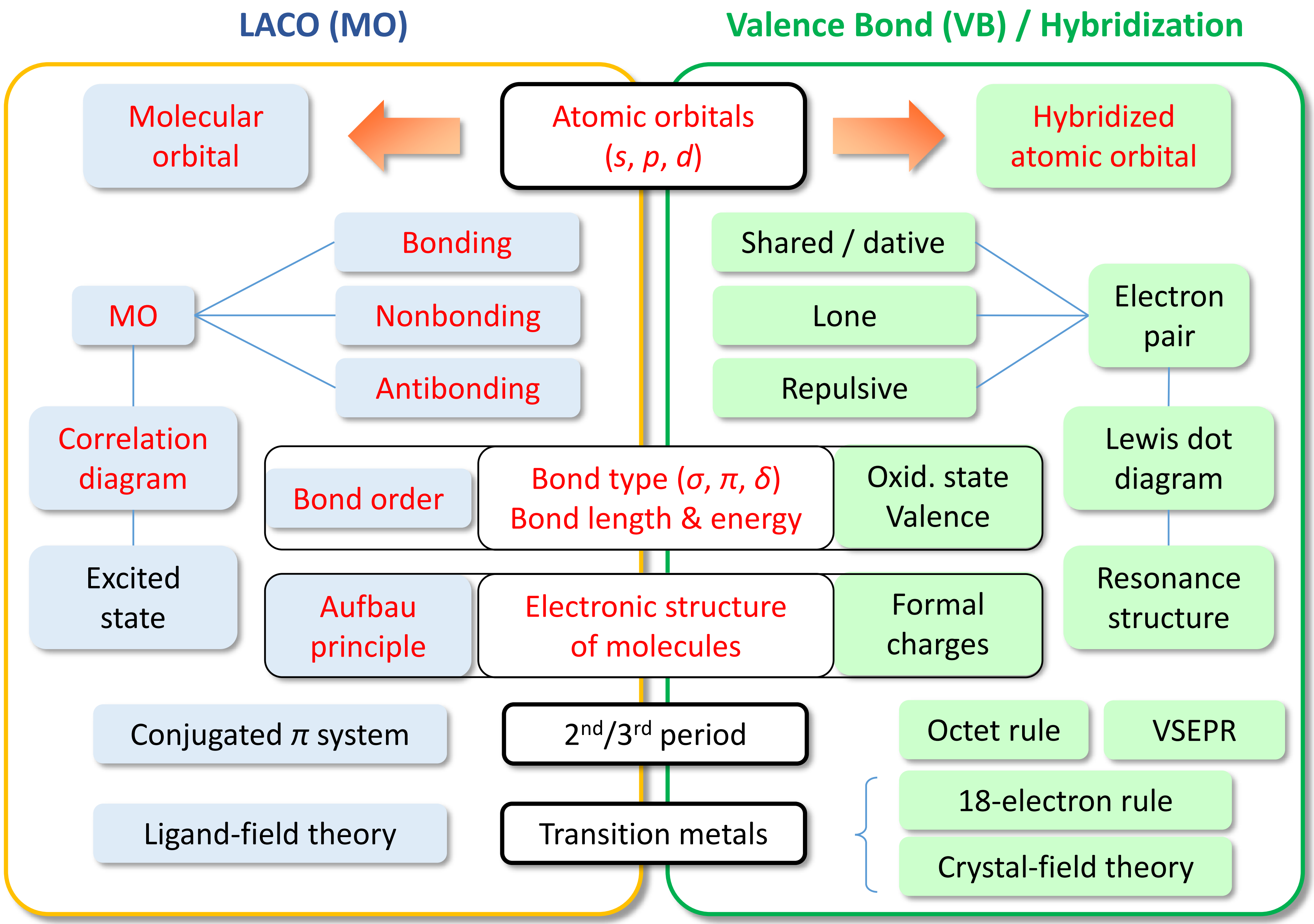
The accuracy of the simple VB wave function can be improved by adding in some ionic character.

Summary and Comparison of the LCAO and Valence Bond Methods

- **LCAO: describe the electronic states of molecules in contexts that require knowledge of energy levels. (Examples include molecular spectroscopy, photochemistry, and phenomena that involve ionization).**
- **VB: describe molecular structure, especially in pictorial ways.**
- **Localized VB σ bonds describe the network holding the molecule together**
- **De-localized LCAO π bonds describe the spread of electron density over the molecule.**

H_2: Levels of Approximation




LACO (MO)
Valence Bond (VB) / Hybridization
Molecular orbital
Atomic orbitals (s, p, d)
Hybridized atomic orbital
Bonding
Nonbonding
Antibonding
MO
Shared / dative
Lone
Repulsive
Electron pair
Correlation diagram
Bond order
Bond type (σ, π, δ)
Bond length & energy
Oxid. state
Valence
Lewis dot diagram
Excited state
Aufbau principle
Electronic structure of molecules
Formal charges
Resonance structure
Conjugated π system
$2^{nd}/3^{rd}$ period
Octet rule
VSEPR
18-electron rule
Ligand-field theory
Transition metals
Crystal-field theory

Lecture 14: Bonding in organic molecules-1

Jin Xie (谢瑨), SPST, ShanghaiTech

Email: xiejin@shanghaitech.edu.cn

Classroom: Teaching Center 401

Wednesday/Friday: 10:15-11:00; 11:10-11:55

2025/11/28

Major points of today's lecture

- **Bonding in organic molecules**

1. **Petroleum Refining and the Hydrocarbons**
2. **The Alkanes**
3. **The Alkenes and Alkynes**
4. **Aromatic Hydrocarbons**
5. **Fullerenes**
6. **Functional Groups and Organic Reactions**

- “数百光年外，一场持续两万多年，涉及整个银河的战争即将结束。获胜的**碳基联邦**为了防止敌人**硅基帝国**死灰复燃，决定以制造横跨数百光年的恒星空白地带来囚禁对手。这也代表着归属这些恒星的行星以及行星上的生命的灭亡。为了银河系中其他大多数的碳基生命，联邦的除星行动冷酷而无慈悲。唯一能逃过这场绝对毁灭的方法就是被发现的生命是否已经具备足够的文明水平并加以证明。”

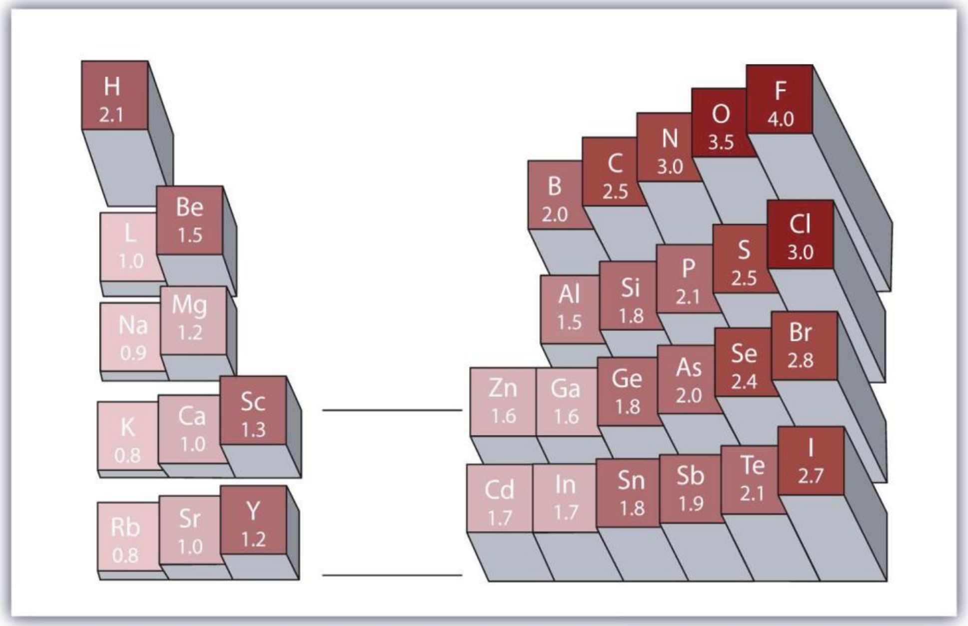
- “临终前，李宝库拼尽最后一丝力量将**力学三大定律**定理篆刻进四个孩子的脑海中。他并不知道，自己的四位学生马上将被选为碳基联邦**文明等级测试**的询问样本。他更不知道，自己刚刚让学生们背下来的力学定律，最终从碳基联邦的除星行动中拯救了地球文明，以及整个太阳系。”

Why Carbon?

Organic Chemistry:
The study of the compounds of carbon.

Average Bond Dissociation Energies (kJ·mol^{-1})

H—H	C—H	C—C	N—N	O—O	F—F	Si—O	P—O
436	414	347	161	146	153	460	377



1. **Small: easily form the double and triple bonds**
2. **Can make four bonds and molecules with different geometries (linear, trigonal, tetrahedral)**
3. **Modest electronegativity: form covalent bonds with elements with both high and low electronegativity**

Petroleum Refining and the Hydrocarbons

When the first oil well was drilled in 1859 near Titusville, Pennsylvania, the future effects of the exploitation of petroleum on everyday life could not have been anticipated.

Major points of today's lecture

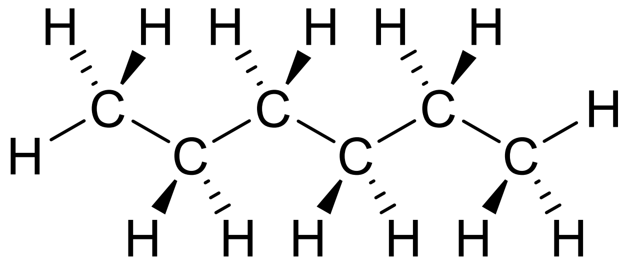
- **Bonding in organic molecules**

1. **Petroleum Refining and the Hydrocarbons**
2. **The Alkanes**
3. **The Alkenes and Alkynes**
4. **Aromatic Hydrocarbons**
5. **Fullerenes**
6. **Functional Groups and Organic Reactions**

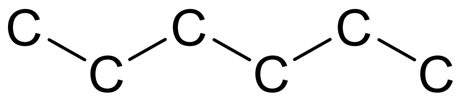
The Alkanes （烷烃）

The most prevalent hydrocarbons in petroleum are the straight-chain alkanes: chains of carbon atoms bonded to one another by single bonds, with enough hydrogen atoms on each carbon atom to bring it to the maximum bonding capacity of four.

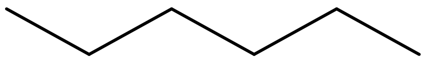
Generic formula C_nH_{2n+2}



$CH_3CH_2CH_2CH_2CH_2CH_3$



$CH_3(CH_2)_4CH_3$, C_6H_{14}

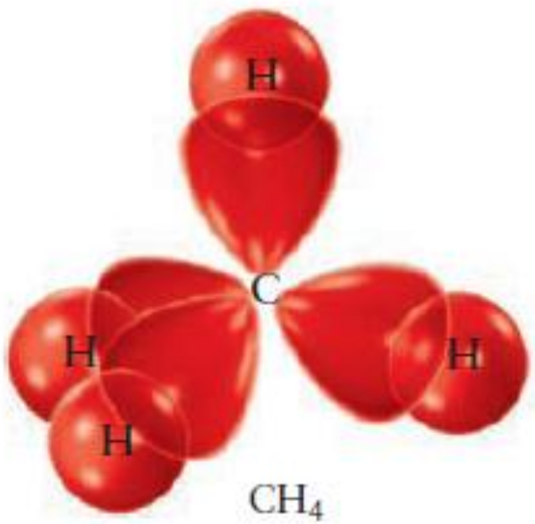


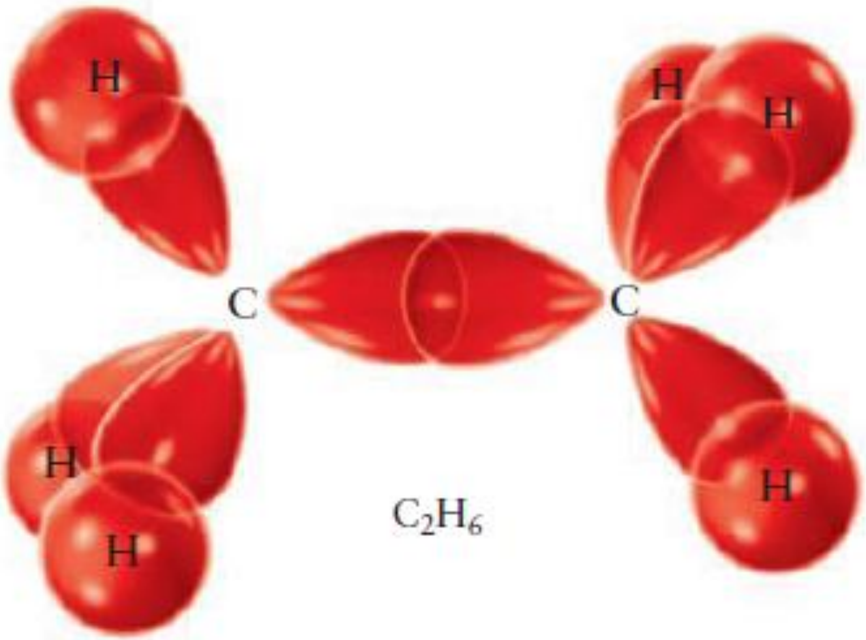
n	Name
1	Methane 甲烷
2	Ethane 乙烷
3	Propane 丙烷
4	Butane 丁烷
5	Pentane 戊烷
6	Hexane 己烷
7	Heptane 庚烷
8	Octane 辛烷
9	Nonane 壬烷
10	Decane 癸烷
11	Undecane 十一烷
12	Dodecane 十二烷
13	Tridecane 十三烷
14	Tetradecane 十四烷
15	Pentadecane 十五烷

The Alkanes

C: $(1s)^2(2s)^2(2p)^2$ sp^3 hybridization

H: $(1s)^1$





The Alkanes

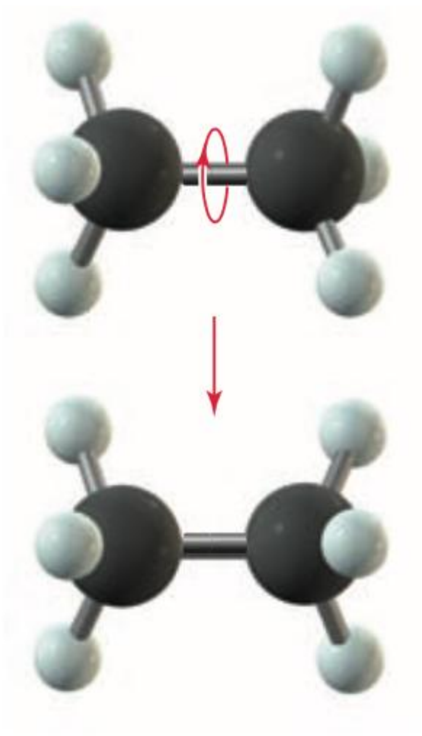
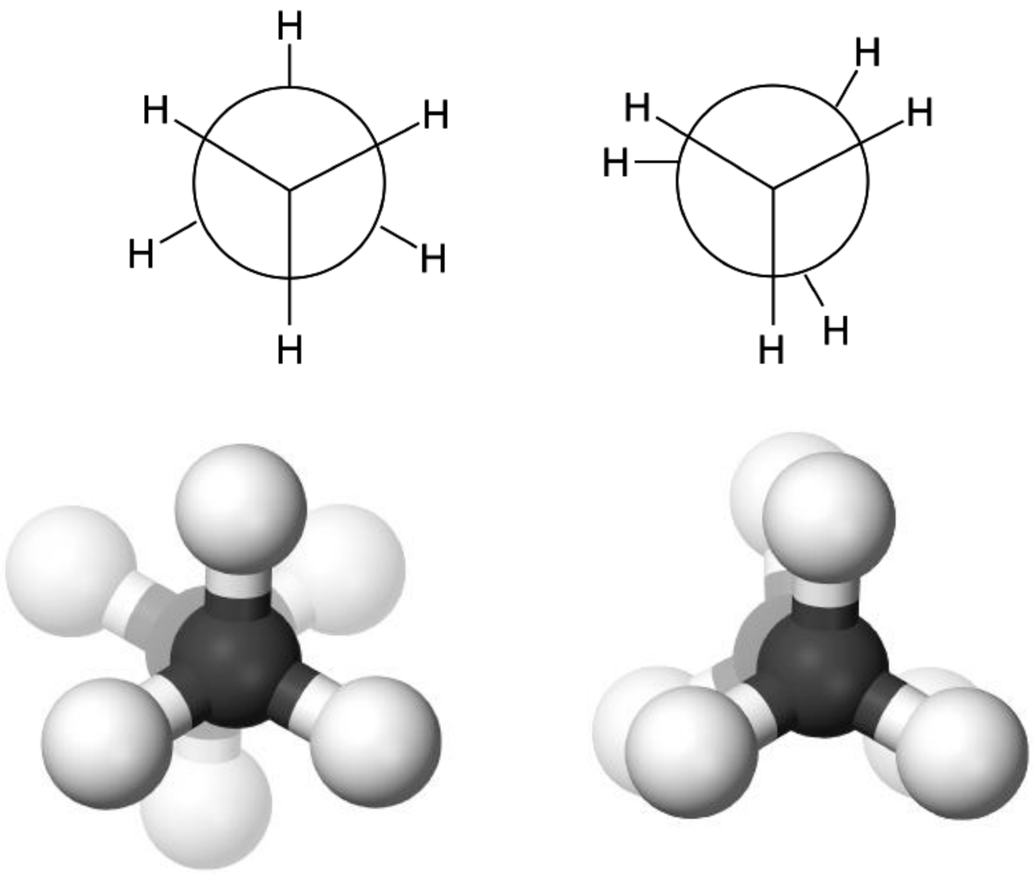


FIGURE 7.2 The two $-CH_3$ groups in ethane rotate easily about the bond that joins them.

different between σ bond and π bond

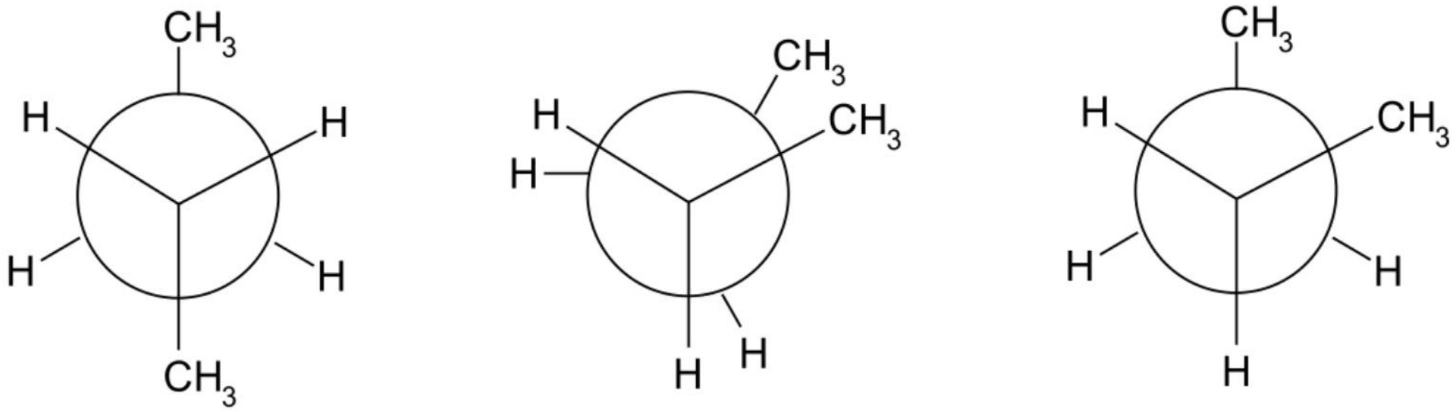
交错式(staggered)　　重叠式(eclipsed)

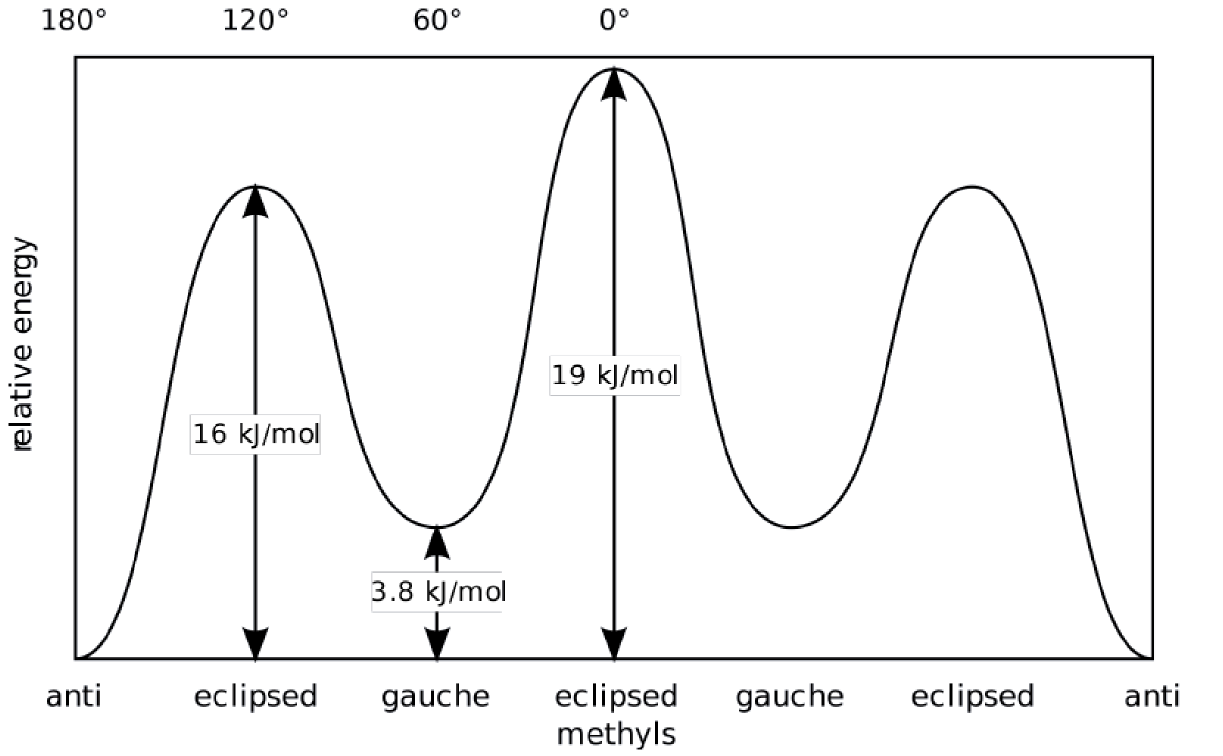


the staggered conformation is more stable by 12.5 kJ/mol than the eclipsed conformation.

The Alkanes

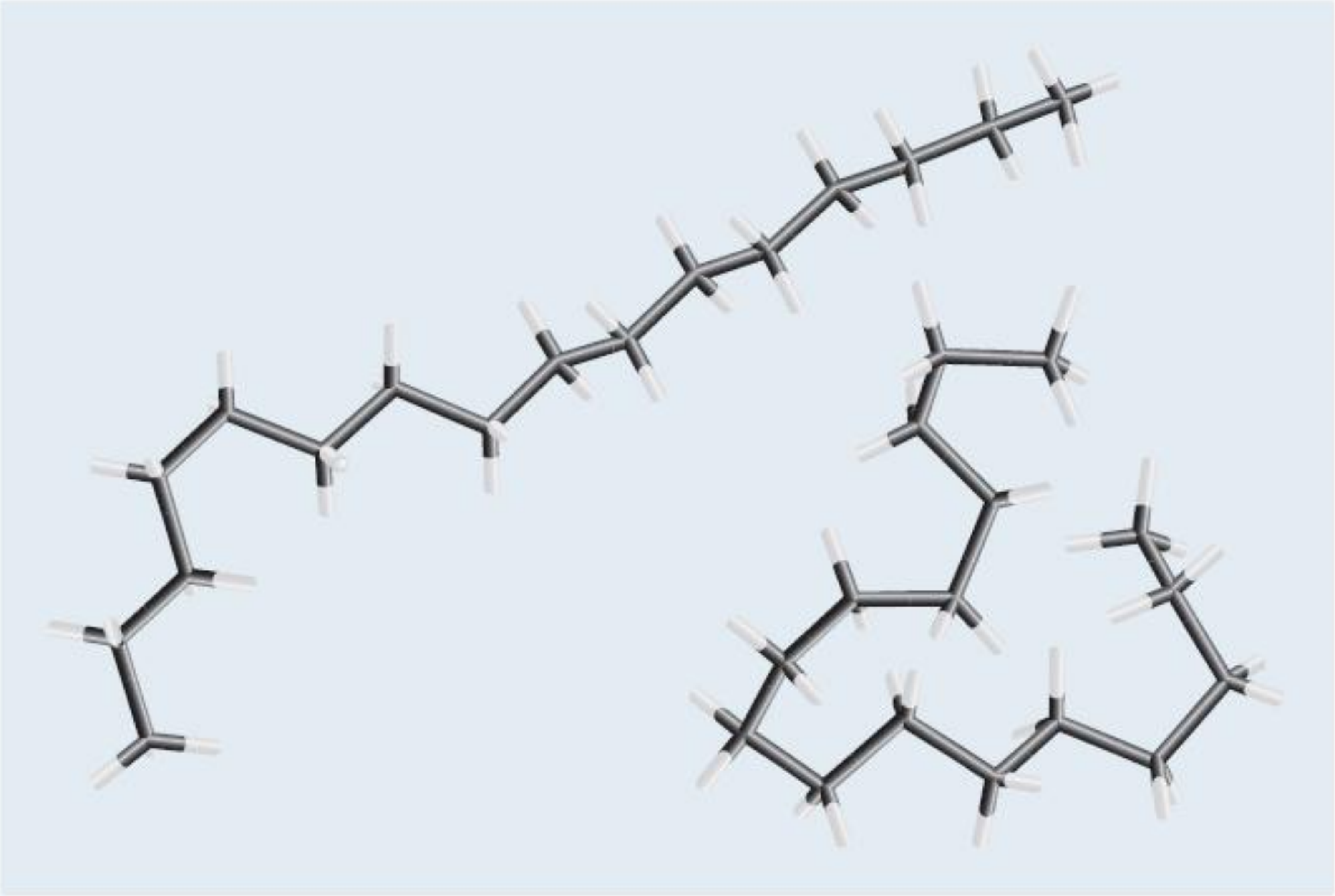
交错式(staggered)、重叠式(eclipsed)与间扭式(gauche)。





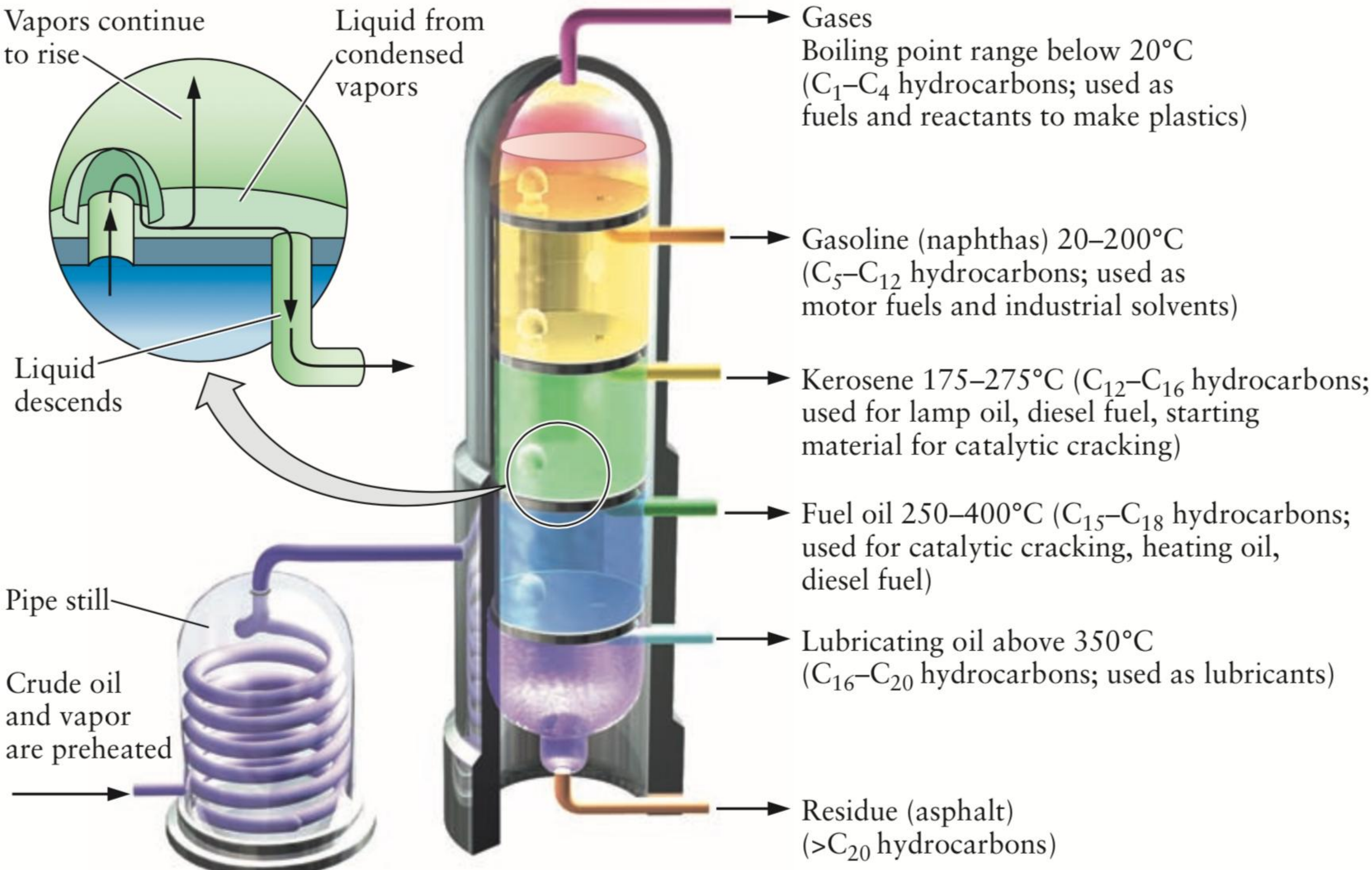
The Alkanes

FIGURE 7.3 Two of the many possible conformations of the alkane $C_{16}H_{34}$. The carbon atoms are not shown explicitly, but they lie at the black intersections. Hydrogen atoms are at the white ends. Eliminating the spheres representing atoms in these tube (or Dreiding) models reveals the conformations more clearly.



Different conformations 构象

Distillation of petroleum



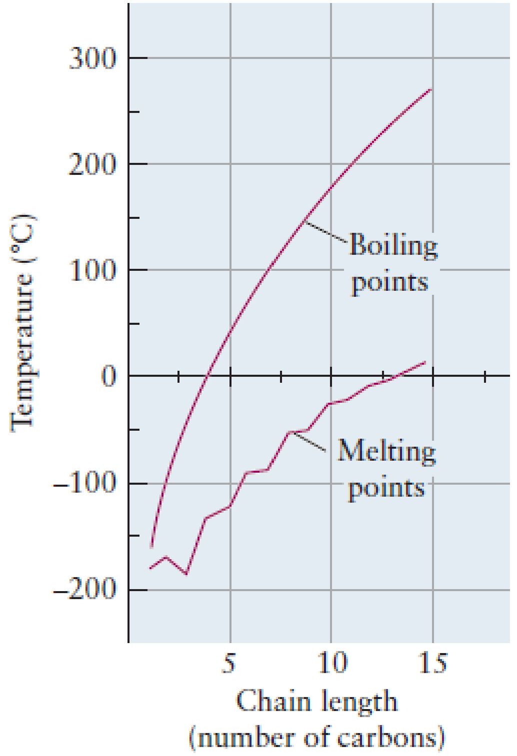
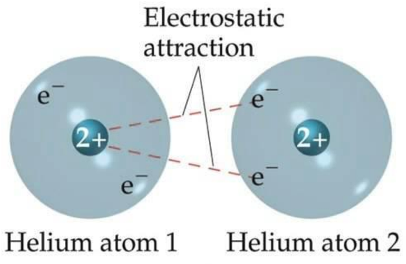
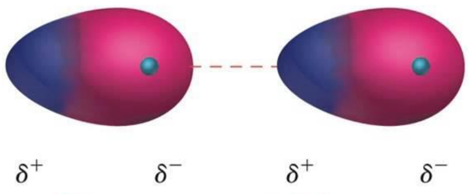


FIGURE 7.4 The melting and boiling points of the straight-chain alkanes increase with chain length *n*. Note the alternation in the melting points: Alkanes with *n* odd tend to have lower melting points because they are more difficult to pack into a crystal lattice.

The Alkanes

It is not a permanent partial charge





Even Distribution of electrons

Temporary UNeven distribution of e- which causes a temporary attraction

伦敦力（London dispersion forces），两个分子可以由于瞬时偶极（两个分子的瞬时偶极总是反向的）间的作用，产生引力，称为伦敦力

increasing strength of dispersion forces between heavier molecules

Butane C_4
B.p.沸点 ~0°C

Diesel C_{10}–C_{22}
B.p. 180–370°C

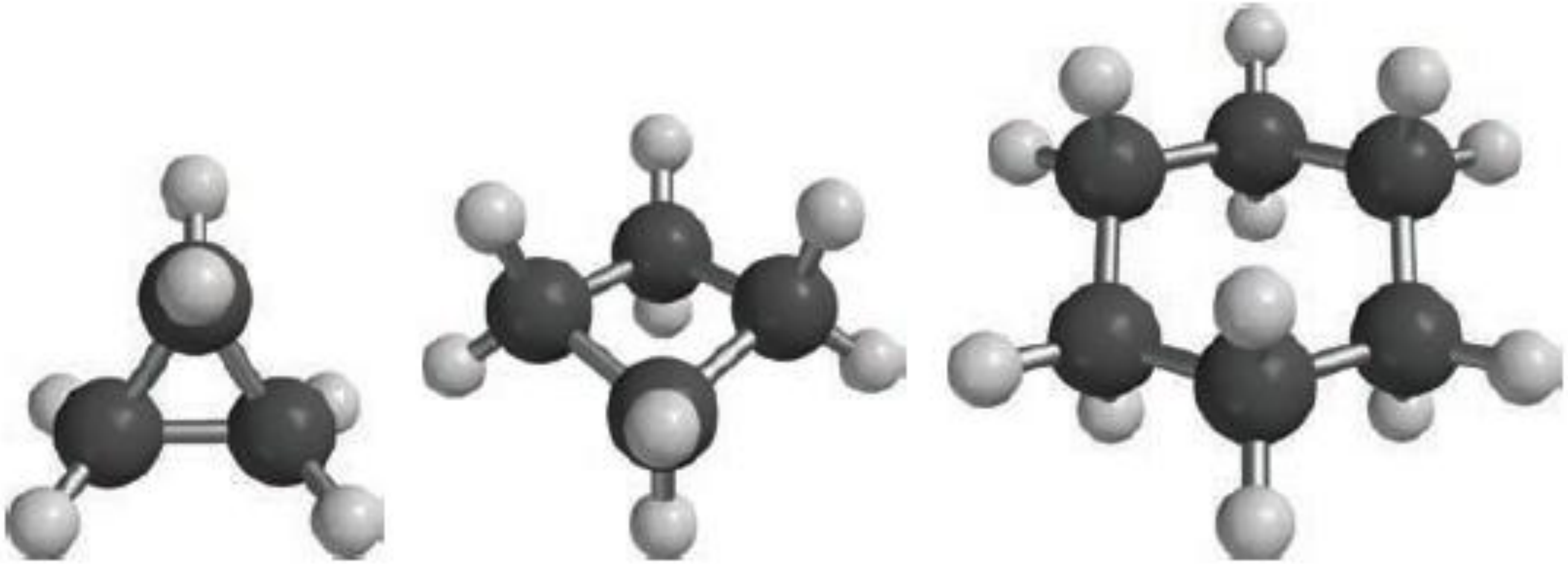
Vaseline C_{20}–C_{30}
M.p. ~37°C

HDPE $C_{\sim 100000}$
M.p. ~130°C

Cyclic Alkanes （环烷烃）

A cycloalkane consists of at least one chain of carbon atoms attached at the ends to form a closed loop

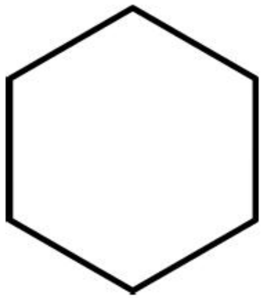
- **cyclo- to the name of the straight-chain alkane**
- **general formula for cycloalkanes having one ring: C_nH_{2n}**

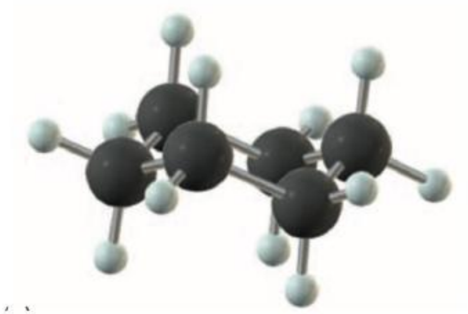


(a) (b) (c)

Cyclic Alkanes

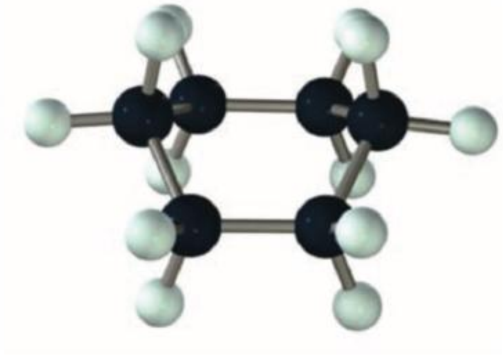
"tetrahedral angle of 109.5 degrees" + "a cyclic structure" = two new structural features (C_6H_{12})





chair conformation

椅型构象



boat conformation

船型构象

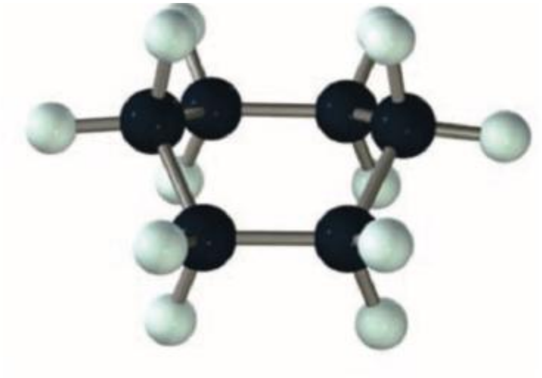
The chair conformation is more stable than the boat: the hydrogen atoms can become quite close and interfere with one another in the boat conformation.

Cyclic Alkanes

When atoms that are not bonded to each other come sufficiently close in space to experience a repulsive interaction, this increase in potential energy reduces the stability of the molecule. Such interactions are called steric strain (空间张力 或 位阻张力)

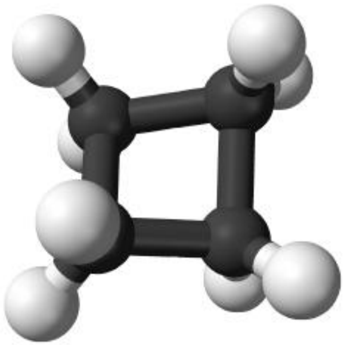
chair conformation

椅型构象



boat conformation

船型构象



This distortion of the bond angle from the tetrahedral value increases the potential energy of the bond above its stable equilibrium value, and the resulting angle strain (角应变)

The Alkanes：geometrical isomers几何异构体

(a)

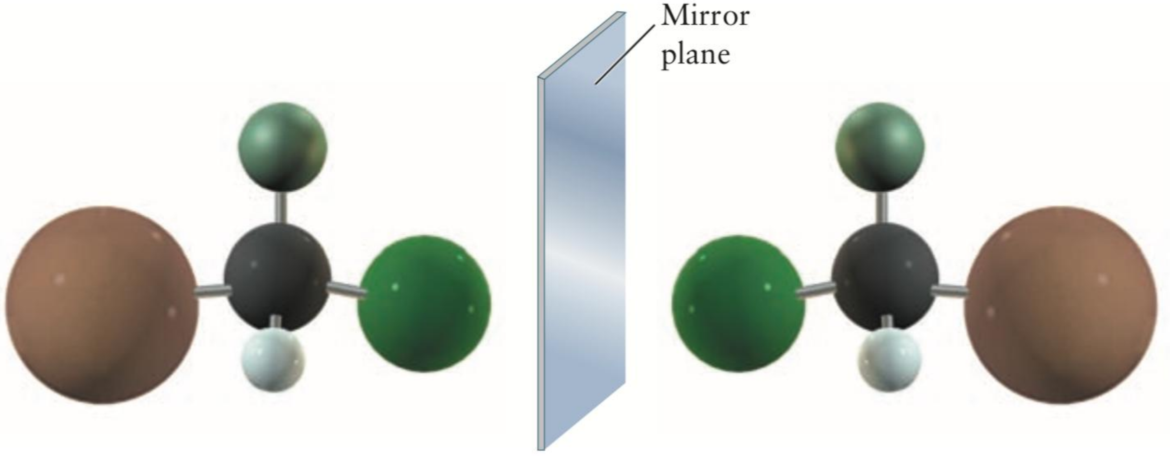
(b)

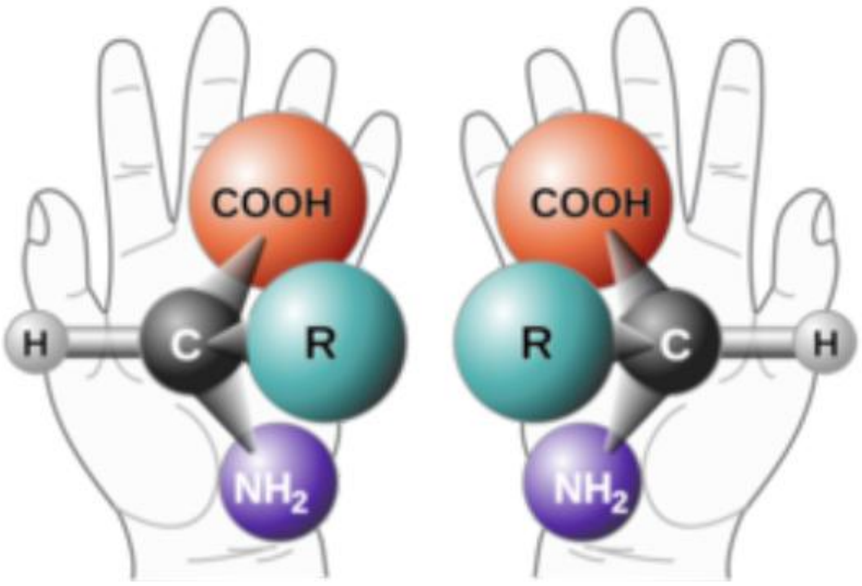
FIGURE 7.8 Two isomeric hydrocarbons with the molecular formula C_4H_{10}. (a) Butane. (b) 2-Methylpropane.

Geometrical isomers几何异构体: same molecular formula but a different bonding structure

The number of possible isomers increases rapidly with increasing numbers of carbon atoms in the hydrocarbon molecule.

The Alkanes：optical isomerism 光学异构体, chirality





Optical isomerism 光学异构体: the two forms rotate the plane of polarized light in different directions; therefore, such molecules are said to be "optically active".

Name the following branched-chain alkane:

$$\begin{array}{ccccc} & CH_3 & & CH_3 & \\ & | & & | & \\ CH_3 & C & CH_2 & C & CH_2CH_3 \\ & | & & | & \\ & H & & CH_2CH_3 & \end{array}$$

Major points of today’s lecture

- **Bonding in organic molecules**

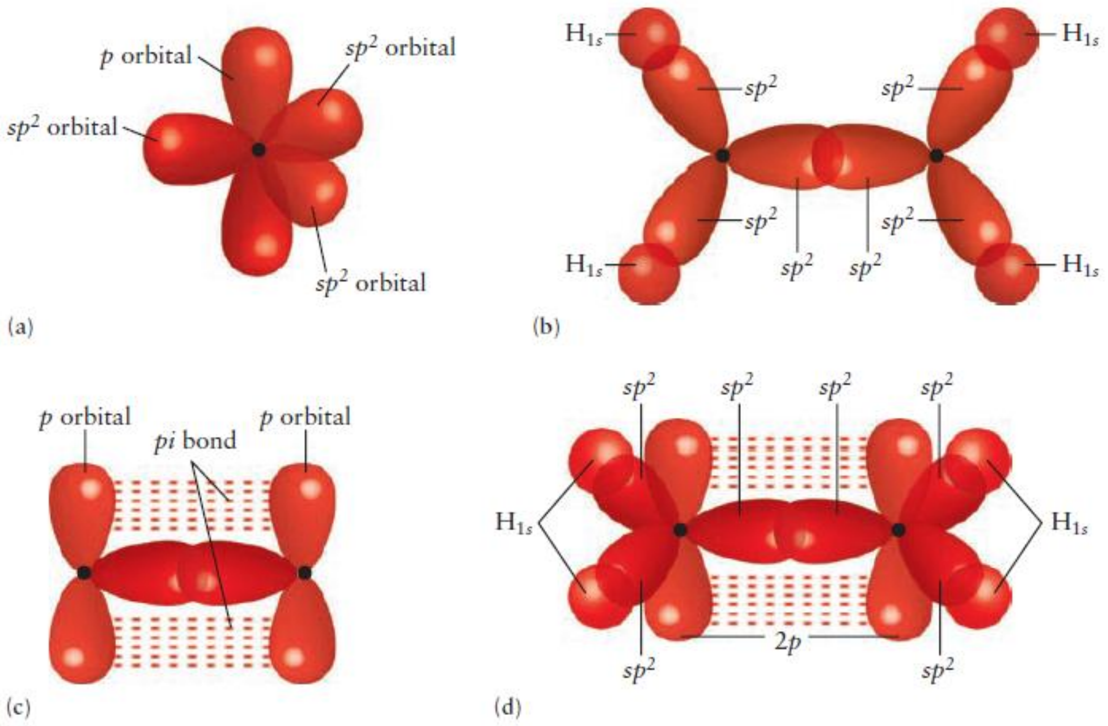
1. Petroleum Refining and the Hydrocarbons
2. The Alkanes
3. The Alkenes and Alkynes
4. Aromatic Hydrocarbons
5. Fullerenes
6. Functional Groups and Organic Reactions

The Alkenes (烯烃) and Alkynes (炔烃)

- **Unsaturated hydrocarbons: have double and triple carbon-carbon bonds**

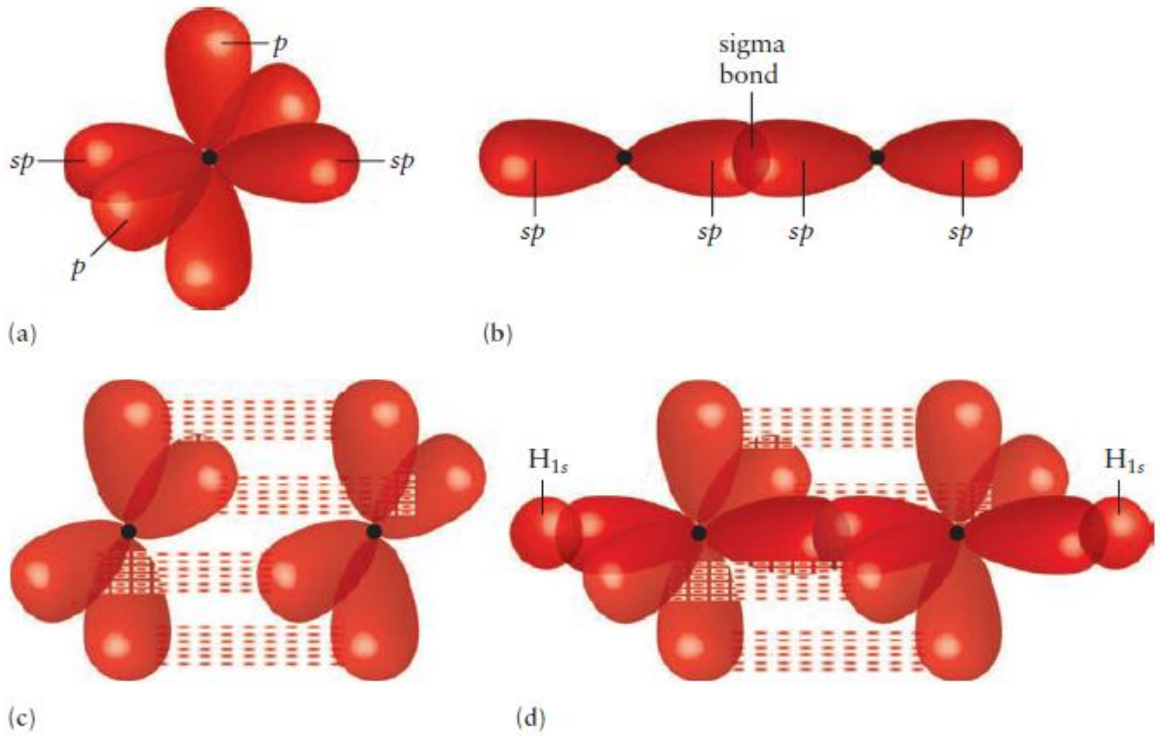
Ethylene (C_2H_4): a double bond between its carbon atoms and is called an alkene.

sp^2 hybridization



Ethylene (C_2H_2): a triple bond between its carbon atoms and is called an alkyne.

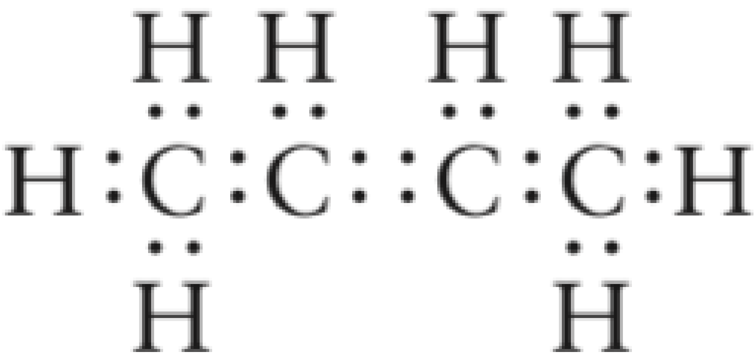
sp hybridization

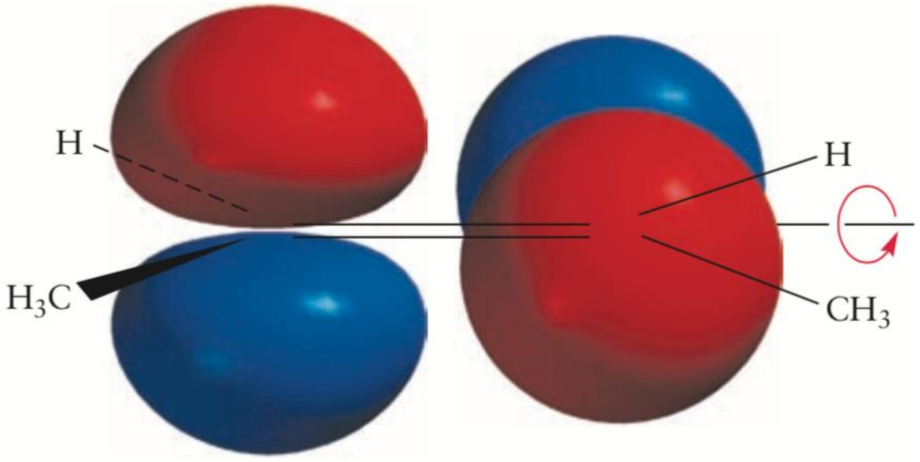


The Alkenes and Alkynes

Localized VB bonds to describe the molecular framework

Delocalized MOs to describe the π electrons





High-energy structure

- **The two outer carbon atoms: SN=4 (sp^3)**
- **The two central carbon atoms: SN=3 (sp^2)**
- **The p orbitals from the two central carbon atoms can be mixed to form a π_z (bonding) MO and a π_z* (antibonding) MO.**

The Alkenes and Alkynes

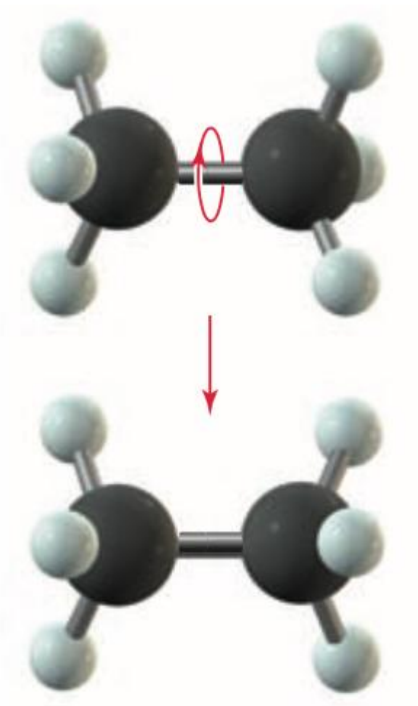
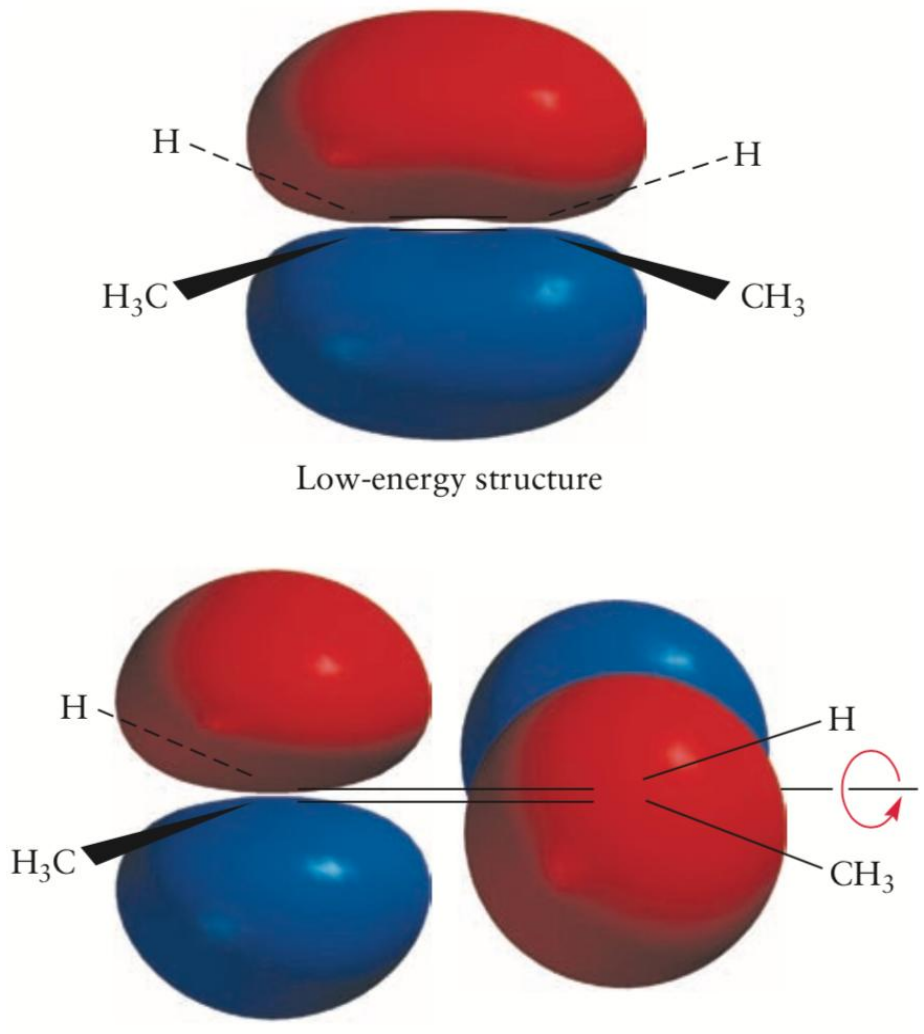


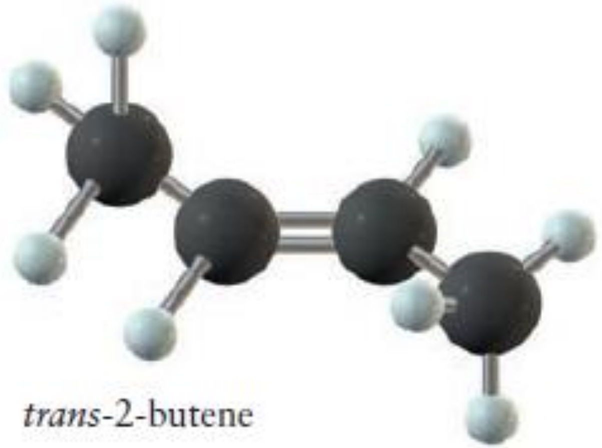
FIGURE 7.2 The two $-CH_3$ groups in ethane rotate easily about the bond that joins them.



different between σ bond and π bond

The Alkenes and Alkynes

cis-2-butene



trans-2-butene

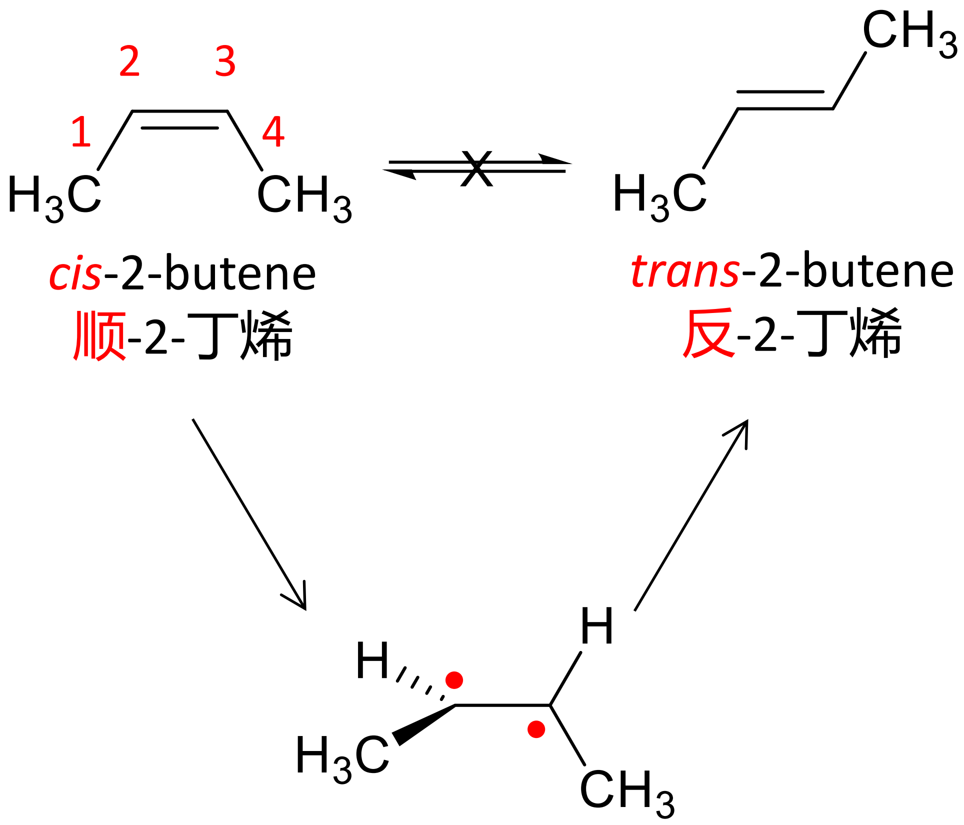
FIGURE 7.14 The two *cis–trans* isomers of 2-butene.

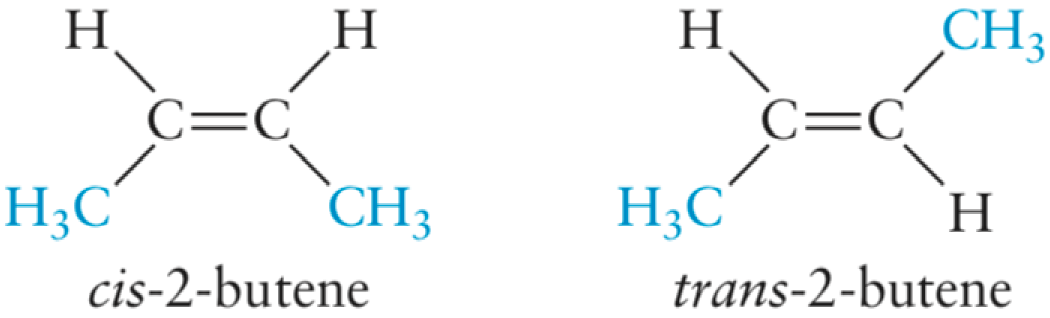
- **Breaking a π bond costs a significant amount of energy: both cis (顺式) and trans (反式) forms are stable, and interconversion between the two is slow.**

- **The change from trans to cis can be accomplished by photochemistry without complete breaking of bonds.**

- **Molecules such as trans-2-butene can absorb ultraviolet light, which excites an electron from a π to a π^* MO.**

The Alkenes and Alkynes





These compounds differ in melting and boiling points, density, and other physical and chemical properties

The Alkenes and Alkynes

Compounds with two double bonds are called dienes (二烯类).

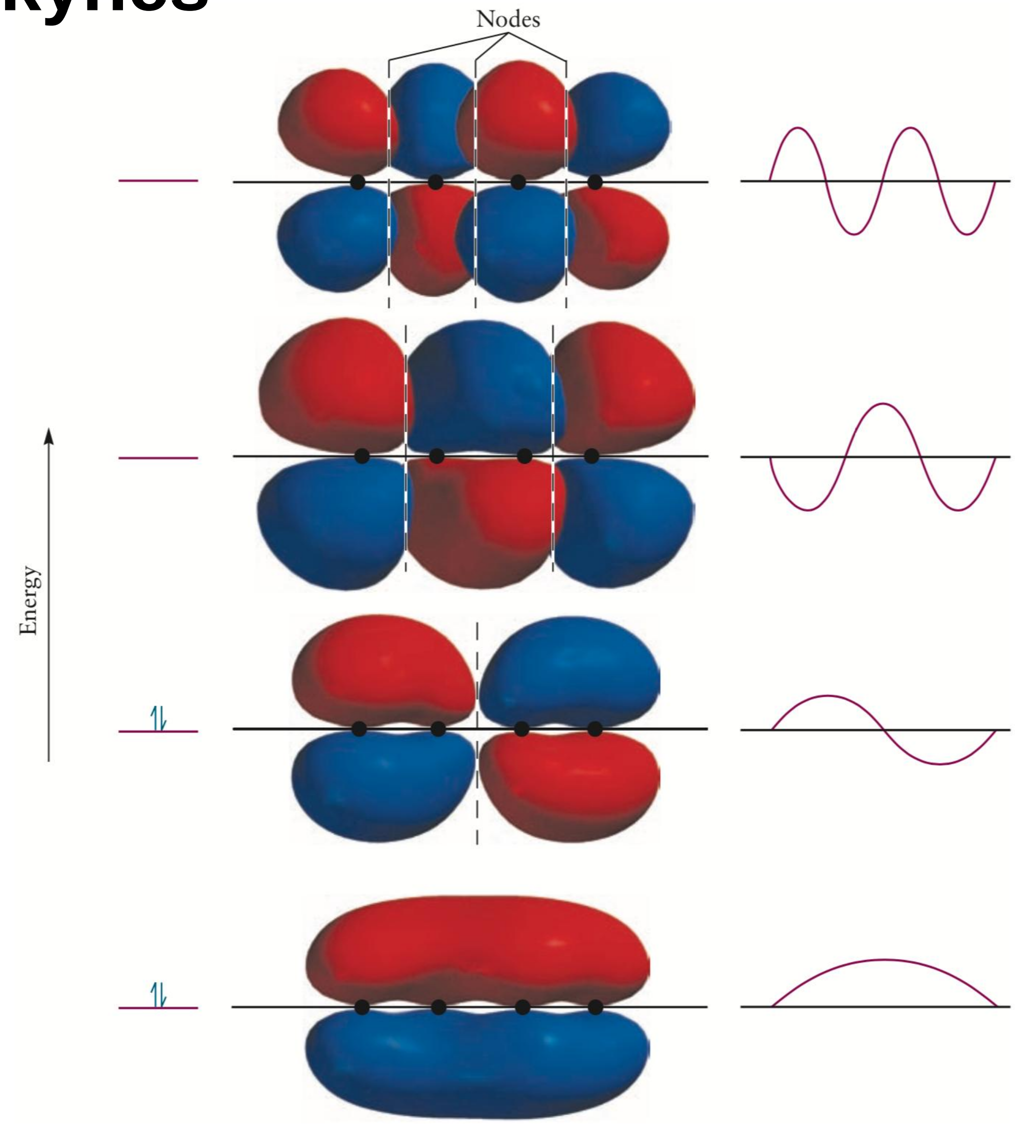
$CH_2{=}CH{-}CH{=}CH{-}CH_3$ **1,3-pentadiene (1,3-戊二烯)**

1,3-butadiene (1,3-丁二烯)

>2 double or triple bonds occur close to each other in a molecule: a delocalized MO.

- **All 4 carbon atoms have SN=3 (sp^2).**
- **The p_z orbitals have maximum overlap when the 4 carbon atoms lie in the same plane (predicted to be planar)**

The Alkenes and Alkynes

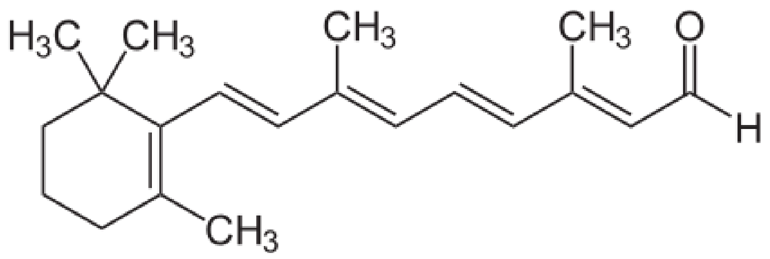


$CH_2{=}CH{-}CH{=}CH{-}CH_3$

one-dimensional particle-in-a-box

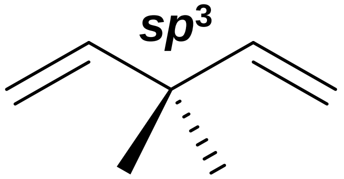
The Alkenes and Alkynes

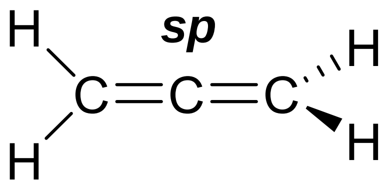
Conjugate system (共轭体系)



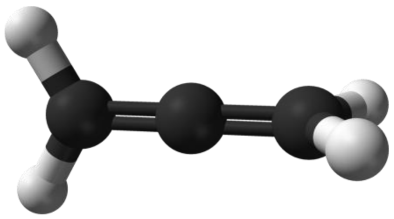
Retinal
视黄醛

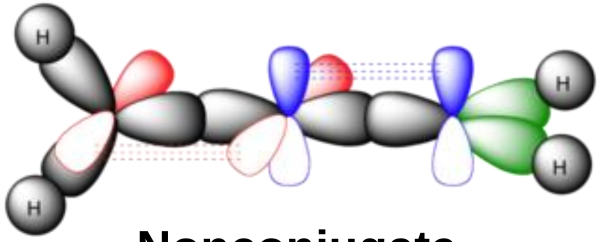
Nonconjugate





Allene 丙二烯



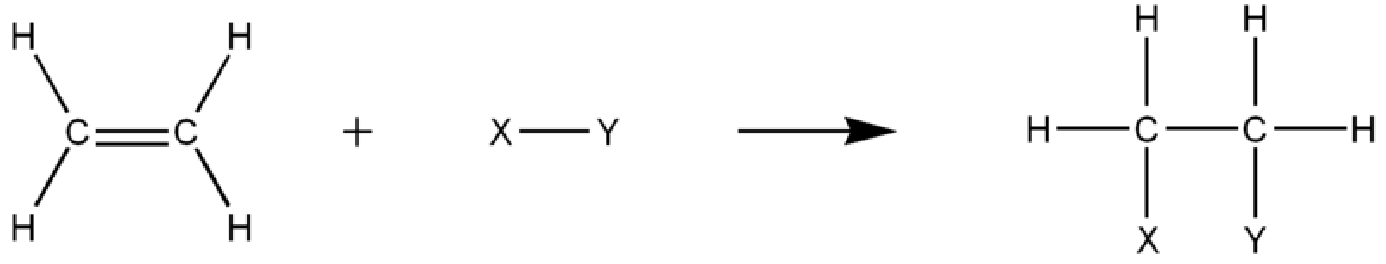


Nonconjugate

The Alkenes and Alkynes

- Alkenes are relatively stable, but are more reactive than alkanes: the reactivity of the C-C pi bond.
- Most reactions of alkenes involve additions to this pi bond, forming new single bonds.

Addition reactions(加成反应)



$$CH_2{=}CH_2 + H_2 \rightarrow CH_3{-}CH_3$$

- Alkenes react in many addition reactions, which occur by opening up the double-bond.
- Most of these addition reactions follow the mechanism of electrophilic addition.

The Alkenes and Alkynes

Hydration

The reaction is catalyzed by strong acids such as sulfuric acid.

$$CH_2{=}CH_2 + H_2O \rightarrow CH_3{-}CH_2OH$$

Halogenation

$$RC{\equiv}CR' + H_2 \rightarrow \textit{cis}\text{-}RCH{=}CR'H$$

$$RC{\equiv}CR' + 2\,Br_2 \rightarrow RCBr_2CR'Br_2$$

Etymology

烃 tīng = 碳 + 氢

hydrocarbon

烷 alkane C_nH_{2n+2}

烯 alkene C_nH_{2n}

炔 alkyne C_nH_{2n-2}

明朝那些事儿

一、太祖派
1、周系 临安王 朱勤烷 (7)
2、楚系 楚庄王 朱孟烷 (2)
3、蜀系 永川王 朱悦烯 (2)
4、岷系 安昌王 朱定烷 (7)
5、唐系 唐靖王 朱琼烃 (2)
6、伊系 伊简王 朱颙炔 (2)
二、成祖派
赵系 赵王 朱慈炔 (12)
三、仁宗派
郑系 郑恭王 朱厚烷 (7)
四、宪宗派
衡系 玉田王 朱厚烃 (7)

秦　王 朱公**锡**永和王 朱慎**镭**

封丘王 朱同**铬**

鲁阳王 朱同**铌**　其子　朱安**汞**

瑞金王 朱在**钠**寿昌王 朱均**铁**

宣宁王 朱成**钴**怀仁王 朱成**钯**

长阳王 朱恩**钠**益阳王 朱恩**铜**

沅陵王 朱恩**铈**长垣王 朱恩**钾**

庆　王 朱帅**锌**弘农王 朱寘**镧**

蒙阴王 朱帅**钾**　韩　王 朱徵**钋**

稷山王 朱效**钛**内丘王 朱效**锂**

唐山王 朱诠**铍**其后　朱效**钯**

新野王 朱弥**镉**伊　王 朱諟**钒**

荥阳王 朱翊**铬**定安王 朱成**鏻**

临安王 朱勤**烷**楚　王 朱孟**烷**

永川王 朱悦**烯**安昌王 朱定**烷**

唐　王 朱琼**烃**伊　王 朱颙**炔**

名字第二个字：一首20字诗，儿子的第几代后人就用诗中的第几个字。

- **太子家是“允文遵祖训，钦武大君胜，顺道宜逢吉，师良善用晟”**
- **燕王家是“高瞻祁见佑，厚载翊常由，慈和怡伯仲，简靖迪先猷”**

第三个字（儿子们统一两个字，后辈三个字），每位朱家后人必须以“五行”为偏旁部首。且从他的儿子起，按照“木火土金水”的顺序循环使用。

The Lewis diagram for propyne (CH_3CCH) is

$$\begin{array}{c}
\mathrm{H} \\
\mathrm{H}:\ddot{\underset{\cdot\cdot}{\mathrm{C}}}-\mathrm{C}\equiv\mathrm{C}:\mathrm{H} \\
\mathrm{H}
\end{array}$$

Discuss its bonding and predict its geometry.

Homework: read “principle of modern chemistry”, chapter 7